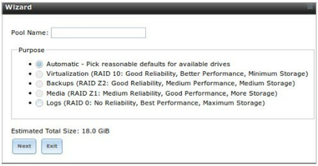

Figure 5: Configuring storage options [Source: freenas.org]

Figure 6: Setting up a private cloud [Source: owncloud.org]

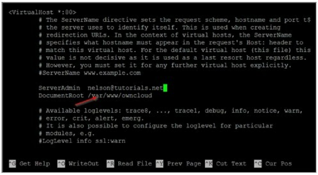

Figure 7: Editing the document root in the configuration files [Source: ittutorials.net]

your storage options depending on your application(s).

Private cloud storage

Another recent trend is cloud storage, given the sudden reduction in free cloud storage offered by providers like Microsoft and Dropbox. Public clouds have multi-tenancy infrastructure and allow for great scalability and flexibility, abstracting away the complexities associated with deploying and maintaining hardware. For instance, the creators of Gluster recently came out with an open source project called Minio to provide this functionality to users. One of the services we will look at is ownCloud, a Dropbox alternative, that offers similar functionality, along with the advantage of being open source.

1. In order to build a private cloud, you require a server running an operating system such as Linux or Windows. ownCloud allows clients to be installed on such a Linux server.

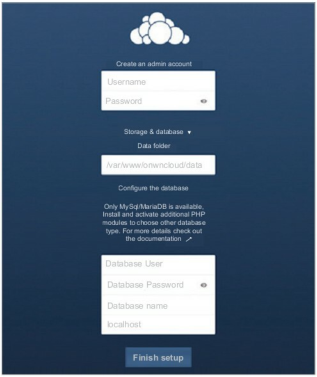

Image 8: Final configuration for ownCloud [Source: ittutorials.net]

2. While installing and running an Apache server on Linux, the *up-load_max_filesize* and *post_max_filesize* flags need to be updated to higher values than the default (2MB).
3. The system is required to have MySQL, PHP (5.4+), Apache, GD and cURL installed before proceeding with the ownCloud installation. Further, a database must be created with privileges granted to a new user.
4. Once the system is set up, proceed with downloading the ownCloud files and extract them to */var/www/ownCloud*.
5. Change the Apache virtual host to point to this ownCloud directory by modifying the document root in */etc/apache2/sites-available/000-default.conf* to */var/www/ownCloud*.
6. Finally, type in the IP address of the server in your browser and you should be able to arrive at the login screen.

While there are trade-offs between cloud-based storage and traditional means of storage, the former is a highly flexible, simplified and secure model of data storage. And with the providers offering more control over deployments, private clouds may well be the main file storage options in the near future! END

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The author has worked with Microsoft Research, CERN and startups in AI and cyber security. An open source enthusiast, he enjoys spending his time organising software development workshops for school and college students. You can connect with him at *https://www.linkedin.com/in/swapneelm* and find out more at *https://github.com/SwapneelM*.